

Nepal Engineering Council Registration Examination
Model Question for Electrical Engineering (AEiE)

Section A (100*1 = 100)

1. Two equal resistances are connected in series across a certain supply. If the resistances are connected in parallel, the power produced will be
a) Two times
b) Four times
c) One-half
d) Equal
2. For transfer of maximum power, the relation between load resistance R_L and internal resistance R_i of the voltage source is
a) $R_L = 2R_i$
b) $R_L = 0.5R_i$
c) $R_L = 1.5R_i$
d) $R_L = R_i$
3. The breakdown voltage of an insulation depends upon... .. value of alternating voltage
a) Average
b) R.M.S
c) Peak
d) Twice the R.M.S
4. At series resonance, the voltage across L or C is
a) Equal to applied voltage
b) Less than applied voltage
c) Much more than applied voltage
d) Equal to voltage across R
5. When the total charge in the capacitor is doubled, the energy stored
a) Remain the same
b) Is doubled
c) Is halved
d) Is quadrupled
6. In a balanced star connected system, line voltage are ahead of their respective phase voltages.
a) 30°
b) 60°
c) 120°
d) 180°
7. Magneto restriction is a phenomenon whereby the magnetism of ferromagnetic material leads to a change in
a) Relative permeability
b) Physical dimension
c) Spontaneous magnetism

- d) Magnetic susceptibility
8. The no load pf of a transformer is small because
- a) Iron loss component of I_0 is large
 - b) Magnetizing component of I_0 is large
 - c) Magnetizing component of I_0 is small
 - d) Iron loss component of I_0 is small
9. The main function of interpoles is to minimize between the brushes and the commutator when the dc machine is loaded
- a) Friction
 - b) Sparking
 - c) Current
 - d) Wear and tear
10. At lagging loads, armature reaction in an alternator is
- a) Cross- magnetizing
 - b) Demagnetizing
 - c) Non effective
 - d) Magnetizing
11. If starting winding of a single-phase induction motor is left in the circuit, it will ...
- a) Draw excessive current and overheat
 - b) Run slower
 - c) Run faster
 - d) Spark at light loads
12. The efficiency of three phase induction motor is approximately proportional to
- a) $1-s$
 - b) s
 - c) N
 - d) Ns
13. Which of the following statements accurately describes the working principles of water turbines?
- a) Water turbines utilize the kinetic energy of water to generate mechanical power.
 - b) Water turbines harness the potential energy of water to produce electricity.
 - c) Water turbines convert the chemical energy of water into rotational motion.
 - d) Water turbines rely on the thermal energy of water to generate hydraulic power.
14. What is a key advantage of diesel power plants in certain applications?
- a) Ability to generate electricity from renewable energy sources
 - b) High efficiency and low fuel consumption
 - c) Can provide electricity during Peak hour
 - d) Minimal environmental impact and emissions
15. Photovoltaic cell or solar cell converts.....
- a) Thermal energy into electricity

- b) Electromagnetic radiation directly into electricity
 - c) Solar radiation into thermal energy
 - d) Solar radiation into kinetic energy
16. Which of the following primary cells has the lowest voltage?
- a) Lithium
 - b) Zinc chloride
 - c) Mercury
 - d) Carbon zinc
17. The measures to improve the transient stability of the power system during the unbalanced or unsymmetrical fault can be taken as a) Single pole switching of circuit breaker (CB)
- b) Excitation control
 - c) Phase shifting transformer
 - d) Increasing turbine valve opening
18. The main exciter used in DC excitation is
- a) Field on stator
 - b) Armature on stator
 - c) Field on rotor
 - d) Field on poles
19. The static error of an instrument implies the
- a) Difference between the measured value and the true value of the quantity
 - b) Accuracy of the instrument
 - c) Error introduced in low varying inputs
 - d) Irreparability of the instrument
20. In a low power factor wattmeter, the compensating coil is connected
- a) in series with the current coil
 - b) in parallel with the current coil
 - c) in series with pressure coil
 - d) in parallel with pressure coil
21. The capacitive transducers are normally used for
- a) Both static and dynamic measurements
 - b) Transient measurements
 - c) Static measurements
 - d) Dynamic measurements
22. In dual slope type of Analog to digital converter, an input hold time is
- a) Almost zero
 - b) Higher than that of flash type Analog to digital converters
 - c) Longest
 - d) Almost one

23. Which of the following is a special purpose register of microprocessor
- a) Program counter
 - b) Instruction register
 - c) Accumulator
 - d) Temporary register
24. If a power transformer has a star-connected primary and a delta-connected secondary, then the Current Transformer (CT) connections on its primary and secondary sides should be
-
- a) Delta and Star respectively
 - b) Star and Delta respectively
 - c) Delta and Delta respectively
 - d) Star and Star respectively
25. In a closed loop control system, with positive value of feedback gain, the overall gain of the system will
- a) Decrease
 - b) Increase
 - c) Be unaffected
 - d) Equal
26. Roots on the imaginary axis makes the system
- a) Stable
 - b) Unstable
 - c) Marginally stable
 - d) Linear
27. Which plots in frequency domain represents the two separate plots of magnitude and phase against frequency in logarithmic value
- a) Polar plots
 - b) Bode plots
 - c) Nyquist plots
 - d) Linear Plots
28. In a thyristor, anode current is made up of
- a) Electrons only
 - b) Electrons or holes
 - c) Electrons and holes
 - d) Holes only
29. The load voltage of a chopper can be controlled by varying the
- a) Duty cycle
 - b) Firing angle
 - c) Reactor position
 - d) Extinction angle

30. HVDC transmission has as compared to HVAC transmission a) Low cost of HVDC transmission
- a) Smaller transformer size
 - b) Smaller conductor size
 - c) Higher corona loss
 - d) Smaller power transfer capabilities
31. A fuse performs:
- a) Detection function only
 - b) Circuit interruption function only
 - c) Both detection and interruption functions
 - d) Circuit connection function only
32. The arc voltage in circuit breaker is
- a) In phase with arc current
 - b) Lagging arc current by 90°
 - c) Leading arc current by 90°
 - d) Lagging arc current by 180°
33. In static overcurrent relay, inverse time characteristics are obtained by
- a) A transistor amplifier
 - b) An integrating circuit
 - c) A transistor switch
 - d) A differentiating circuit
34. Surge absorber is used for protection against
- a) High voltage low frequency oscillation
 - b) High voltage high frequency oscillation
 - c) Low voltage high frequency oscillation
 - d) Low voltage low frequency oscillation
35. Resistance earthing is employed for voltage between
- a) 3.3kV and 11kV
 - b) 11kV and 33kV
 - c) 33kV and 66kV
 - d) 66kV and 132kV
36. Shunt compensation in the high voltage line is used to improve
- a) Stability and fault level
 - b) Fault level and voltage profile
 - c) Voltage profile and stability
 - d) Stability, fault level and voltage profile
37. The main drawback of overhead system over underground system is
- a) Underground system is more flexible than overhead system
 - b) Higher charging current

- c) Surge problem
 - d) High initial cost
38. Which gas has the highest breakdown strength at atmospheric pressure? a) Air
b) Nitrogen
c) SF₆
d) Oxygen
39. If the length of the cross arm is increased, the string efficiency
a) Becomes zero
b) Increases
c) Remains unaffected
d) Decreases
40. Practical loads are usually
a) Resistive
b) Capacitive
c) Inductive
d) Resistive + Inductive
41. The feeder is designed mainly from the point of view of
a) Its current carrying capacity
b) Voltage drop in it
c) Operating voltage
d) Operating frequency
42. Synchronous condenser used for the power factor improvement is synchronous motor which operates at
a) No load with leading current
b) Full load with lagging current
c) No load with lagging current
d) Full load with leading current
43. The gas used in gas filled filament lamp is
a) Helium
b) Oxygen
c) Nitrogen
d) Ozone
44. Active power and apparent power are represented by
a) kW and kVAR
b) kVAR and kVA
c) kVA and kVAR
d) kW and kVA
45. The typical active load is

- a) Hoist
 - b) Blower
 - c) Pump
 - d) lathe
46. For heating plywood, the frequency should be
- a) (1-2) MHZ
 - b) (10-25) KHZ
 - c) 1KHZ
 - d) 100HZ
47. In Kando System of track electrification
- a) Single phase AC is converted to DC
 - b) Single phase AC is converted to 3 Phase AC
 - c) 3 phase AC is converted into DC
 - d) 3 phase AC is converted into Single Phase AC
48. Two Part tariff is charged on the basis of
- a) Connected load
 - b) Unit consumed
 - c) Maximum demand
 - d) Connected load and Unit consumed
49. The chances of occurrence of corona are maximum during
- a) Humid weather
 - b) Dry weather
 - c) Winter
 - d) Hot summer
50. For a transmission line which among the following relation is true?
- a) $-AB + CD = 1$
 - b) $AD + BC = 1$
 - c) $AB - CD = -1$
 - d) $-AD + BC = 1$
51. If all the sequence voltages at the fault point in a power system are equal, then the fault is a
- a) Three phase fault
 - b) Line to ground fault
 - c) Line to line fault
 - d) Double line to ground fault
52. In load flow studies of a power system, a voltage control bus is specified by
- a) Real power and reactive power
 - b) Reactive power and voltage magnitude
 - c) Voltage and voltage phase angle

- d) Real power and voltage magnitude
53. Which stability information is obtained from the equal area criterion?
- Absolute stability
 - Transient stability
 - Steady state stability
 - Transient stability and Steady state stability
54. VAR compensators are used to
- Increase the active power
 - Increase the reactive power
 - Decrease the active power
 - Decrease the reactive power
55. Standard dimensions (mm x mm) of A3 drawing sheet in mm is
- 11.69×16.54
 - 29.7×42
 - 297×420
 - 420×280
56. Which of the following methods of charging depreciation of an asset has increased amount of depreciation as the age of asset increases
- Sum-of-year digit
 - Sinking fund
 - Diminishing balance
 - Straight line
57. The process of optimizing the project's limited resources without extending the project duration is known as
- project crashing
 - resource levelling
 - resource smoothing
 - networking
58. The process of composing/raising the required fund from different sources such as equity, preferred stock, bond and debenture is known as
- capital structure planning
 - project financing
 - capital budgeting decision
 - deducing earning per share
59. In which of the following society people used to seek their existence on growing plants for their cattle and domestic animals
- pastoral society
 - tribal society
 - horticultural society
 - agricultural society
60. According to Nepal Engineering Council Act, 2055 (with revised, 2080), all engineering academic institutions shall be in the Council.
- affiliated
 - united
 - recognized

d) associated

61. If temperature coefficient of resistance is α_0 at temperature 0°C , then at temperature $t^\circ\text{C}$ the temperature coefficient of resistance α_t is:

- a) $\alpha_t = 1/(1/\alpha_0 + t)$
- b) $\alpha_t = \alpha_0/(1/\alpha_0 + t)$
- c) $\alpha_t = 1/(1 + \alpha_0.t)$
- d) $\alpha_t = 1/(1 + t/\alpha_0)$

62. Two voltages across two series connected circuit elements are $v_1 = 50 \sin \omega t$ V and $v_2 = 30 \sin (\omega t - 30^\circ)$ V. The phase angle of the applied rms voltage with respect to v_1 is:

- a) -30°
- b) $+30^\circ$
- c) -11.16°
- d) $+16.16^\circ$

63. Transient current in an R-L-C circuit is oscillatory when

- a) $R = 0$
- b) $R = 2\sqrt{\frac{L}{C}}$
- c) $R > 2\sqrt{\frac{L}{C}}$
- d) $R < 2\sqrt{\frac{L}{C}}$

64. In a series resonance circuit, the Q-factor is:

- a) $\frac{1}{R\sqrt{LC}}$
- b) $\frac{1}{R}\sqrt{\frac{C}{L}}$
- c) $\frac{1}{R}\sqrt{LC}$
- d) $\frac{1}{R}\sqrt{\frac{L}{C}}$

65. The kVA rating of an ordinary 2-winding transformer is increased when connected as an autotransformer because

- a) Transformer ratio is increased
- b) Secondary voltage is increased
- c) Energy transferred both inductively and conductively
- d) Secondary current is increased

66. Critical resistance of dc generator is proportional to:

- a) N
- b) $1/N$
- c) N_2
- d) $1/N_2$

67. A 480 MW 3-phase Y-connected synchronous generator has a rated voltage of 20 kV at 0.8 pf lagging. The line current when operating at full load rated condition is:
- a) 13.3 kA
 - b) 17.3 kA
 - c) 23.25 kA
 - d) -27.36 kA
68. The effect of increasing the length of air-gap in an induction motor will be to increase the
- a) Power factor
 - b) Magnetizing current
 - c) Speed
 - d) Air-gap flux
69. Hydro-graph is similar to:
- a) Mass curve
 - b) Volume curve
 - c) Energy load curve
 - d) Load curve
70. Betz law finds application in
- a) Geothermal
 - b) Solar
 - c) MHD
 - d) Wind mills
71. The ratio of the area under the load curve to the total area under the rectangle in which it is contained gives the value of:
- a) Utilization factor
 - b) Demand factor
 - c) Diversity factor
 - d) Load factor
72. A four-stroke diesel engine has:
- a) Four valves
 - b) Three valves
 - c) Two valves
 - d) One valve
73. The error occurred in a measurement system due to human mistake is:
- a) Static
 - b) Dynamic
 - c) Gross
 - d) Magnetic

74. For a balanced three phase three wire system with input power of 10000 W at 0.9 power factor, the readings on both wattmeter is:
- a) 3650 W, 6350 W
 - b) 5000 W, 5000 W
 - c) 7600 W, 1200 W
 - d) 7000 W, 3000 W
75. Which is not an analog-to-digital (ADC) conversion error?
- a) Differential non-linearity
 - b) Missing code
 - c) Incorrect code
 - d) Offset
76. A PT is a device which is:
- a) Electro-statically coupled
 - b) Electrically coupled
 - c) Conductively coupled
 - d) Electromagnetically coupled
77. A control system working under unknown random actions is called:
- a) Adaptive control system
 - b) Stochastic control system
 - c) Computer control system
 - d) Digital data system
78. With negative feedback in a closed loop control system, the system sensitivity to parameter variation:
- a) Becomes infinite
 - b) Becomes zero
 - c) Decreases
 - d) Increases
79. **An SCR is turned off by:**
- a) Reducing anode voltage to zero
 - b) Reducing cathode voltage to zero
 - c) Reducing gate voltage to zero
 - d) Reverse biasing the gate
80. A three-phase full controlled converter can operate as a:
- a) Converter for $\alpha = 0$ to 120°
 - b) Converter for $\alpha = 0$ to 90°
 - c) Converter for $\alpha = 0$ to 60°
 - d) Converter for $\alpha = 0$ to 180°

81. A fuse wire of circular cross section has a radius of 0.8 mm. The wire blows off at a current of 8 A. What will be the radius of the wire that will blow off at a current of 1 A?
- a) 0.1 mm
 - b) 0.2 mm
 - c) 0.4 mm
 - d) 0.8 mm
82. The current chopping tendency is minimized by using SF₆ gas at relatively
- a) Low pressure and low velocity
 - b) Low pressure and high velocity
 - c) High pressure and high velocity
 - d) High pressure and low velocity
83. Circuit breaker specifications are given as 1500 A, 1000 MVA, 33 kV, 3 seconds, and three-phase oil circuit breaker. What is its breaking capacity?
- a) 3000 MVA
 - b) 2000 MVA
 - c) 1500 MVA
 - d) 1000 MVA
84. Which of the following devices will receive voltage surge first travelling on the transmission line?
- a) Lightning arrestors
 - b) Relays
 - c) Stepdown transformer
 - d) Switchgear
85. The existing highest voltage transmission voltage in Nepal is:
- a) 220 kV
 - b) 132 kV
 - c) 400 kV
 - d) 500 kV
86. Earth wire is made of:
- a) Copper
 - b) Aluminium
 - c) Iron
 - d) Galvanised steel
87. In an overhead transmission line if the supports are at equal levels and tension is increased two times, then:
- a) Sag also increases to two times
 - b) Sag becomes zero
 - c) Sag decreases to the half of the previous value
 - d) Sag also increases to four times

88. Series capacitors can be used in distribution lines:
- a) To provide reactive power compensation
 - b) To reduce the receiving-end voltage under light load conditions
 - c) To reduce the voltage drop
 - d) To reduce line losses
89. What is the maximum load that can be connected in a circuit connecting only lighting points?
- a) 500 Watts
 - b) 750 Watts
 - c) 800 Watts
 - d) 1000 Watts
90. A lighting sub-circuit has 100 W lamps. If the rated current of the fused in this circuit is 5 A, the maximum number of light points on this circuit without violating regulations and safety should be:
- a) 8
 - b) 9
 - c) 14
 - d) 16
91. If the speed of dc shunt motor increases, the back emf:
- a) Increases
 - b) Decreases
 - c) Remains constant
 - d) First increases and then decreases
92. The armature torque of a dc motor depends on its:
- a) Field flux alone
 - b) Speed alone
 - c) Armature current alone
 - d) Both field flux and armature current
93. The load must not be connected directly on:
- a) Transmission line
 - b) Distributor
 - c) Service mains
 - d) LT 230 V line
94. For newton-raphson load flow analysis, the transmission lines are represented by:
- a) Nominal T
 - b) Nominal pi
 - c) Tunned power network
 - d) Ladder network

95. For a single line to ground (SLG) fault the zero sequence current is given by j 3.0 pu. The current carried by the neutral during the fault is:
- a) j 1.0 pu
 - b) j 3.0 pu
 - c) j 6.0 pu
 - d) j 9.0 pu
96. Due to proximity effect, the increase in conductor resistance is not negligible in:
- a) Communication line
 - b) Optical cable line
 - c) Overhead transmission line
 - d) Underground cable
97. Which of the following is a combination of T-square, protractor and scales? Which helps to draw at a faster rate on following?
- a) Fixing clamp
 - b) Clutch spring
 - c) Erasing shield
 - d) Mini drafter
98. What is the dimension of A1 size drawing sheet?
- a) 594 mm x 841 mm
 - b) 1189 mm x 841 mm
 - c) 1230 mm x 880 mm
 - d) 148 mm x 210 mm
99. A project takes 20 days along the critical path and has standard deviation of 4 days. How many days does it take to complete the project?
- a) 24 days
 - b) 20 days
 - c) 18 days
 - d) 16 days
100. According to Nepal Engineering Council Act, 2055, what should be the minimum qualification of Chairperson of Council ?
- a) Bachelor's degree in engineering and at least fifteen years of experience in the engineering profession
 - b) Bachelor's degree in engineering and at least ten years of experience in the engineering profession
 - c) Bachelor's degree in engineering
 - d) Master's degree in engineering

